CS-550 PA-1
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Performance Evaluation

3. Measuring how the transfer speed changes when varying the number of concurrent clients (N). All the clients are requested to download files from the server concurrently. Each client should create at least 10 threads downing a fixed number (set by students) of files simultaneously. and each client needs to repeat the experiments for a fixed number (set by students).

a. N can be 2, 4, 8, 16

b. Graph your results

- Performance evaluation for problem 3:
- Number of clients connected concurrently = N
- Run for N = 2, 4, 8, 16
- Each client = 10 threads
- Download file size = 8kb (this can be changed through the script)
- Total number of experiments executed = 5 (can be configured using "numExperiments" in the *config.properties* file.)
- Results:

  For N=2:

Client 1	2812.85 ns
Client 2	2718.23 ns

For N=4:

Client 1	6317.418
Client 2	6401.05
Client 3	6300.902
Client 4	6603.41

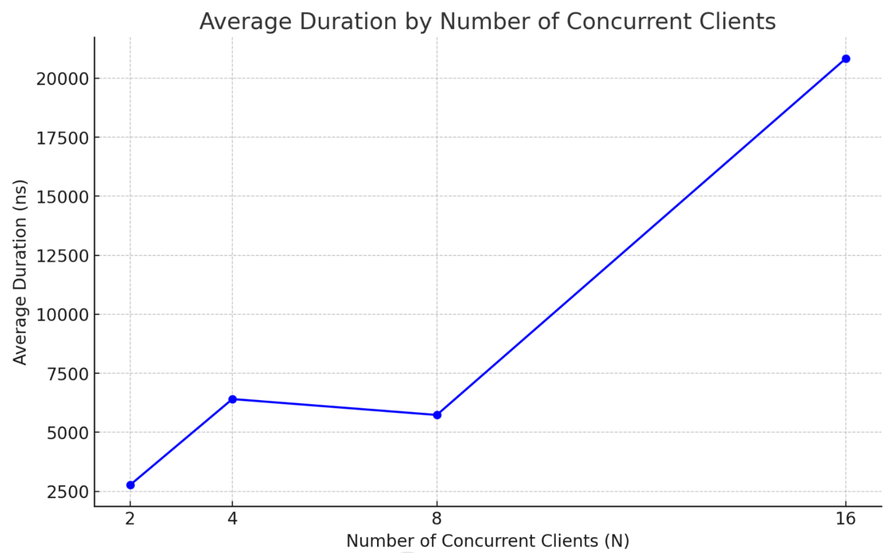
For N=8:

Client 1	7701.93 ns
Client 2	7766.33 ns
Client 3	7774.94 ns
Client 4	8292.83 ns
Client 5	2387.78 ns
Client 6	4468.60 ns
Client 7	3838.22 ns
Client 8	3622.58 ns

For N=16:

Client 1	6977.376 ns
Client 2	2851.434 ns
Client 3	2896.66 ns
Client 4	3808.208 ns

Client 5	32492.824 ns
Client 6	30998.1 ns
Client 7	3974 ns
Client 8	5128.81 ns
Client 9	4565.516 ns
Client 10	48272 ns
Client 11	60428.9 ns
Client 12	62671.292 ns
Client 13	16838.286 ns
Client 14	8034.35 ns
Client 15	8650.184 ns
Client 16	34755.794 ns



4. Measuring how the transfer speed changes when varying the download file size.
    a. Different file sizes are 128, 512, 2k, 8k, 32k
    b. For each size of file in a), performing the following test
        • 4 clients concurrently request to download files from the server. Each client creates at least 10 threads to download a fixed number (set by students) of files simultaneously, and each client needs to repeat the experiments for a fixed number (set by students).
    c. Graph your results

- Performance evaluation for problem 4
- Total number of clients = 4
- Each client = 10 threads
- Number of experiments = 5 (configurable through "numExperiments")
- Following are the execution results of downloading each of the given file sizes:

File Size	Average Duration (ns)
128b	4893.908
512b	5318.128
2kb	5290.1345
8kb	5037.8265
32kb	6485.567